

Miguel

Miguel: Your Intelligent Cycling Companion

Your ride. Your passion. My expertise. 🚴

From the professional peloton to your garage,
everything cycling under one roof.

Gemini Notebook





THE KNOWLEDGE FRAGMENTATION PROBLEM

Cycling is an ecosystem of massive complexity—spanning biomechanics, mechanical engineering, nutrition science, and rich sporting history.

THE GARAGE

Mechanical troubleshooting is buried in conflicting 20-minute YouTube tutorials.

Information Access: 35%

THE BODY

Training and nutrition science is locked behind expensive coaching or generic web searches.

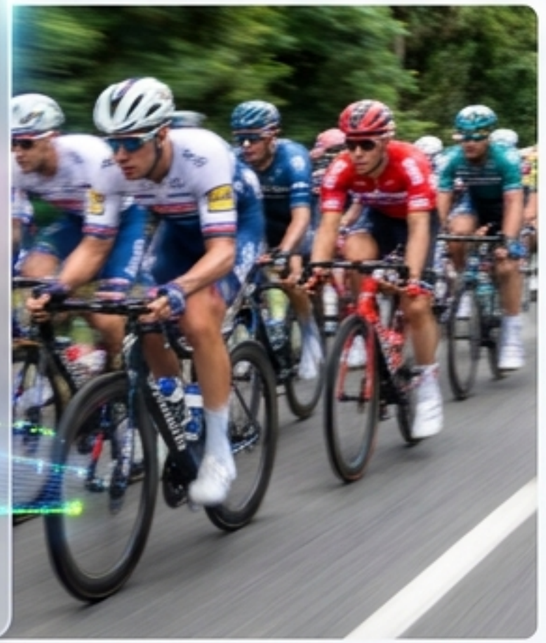
Personalized Guidance: 20%

THE SPORT

Pro racing tactics and culture are scattered across outdated forums and paywalled articles.

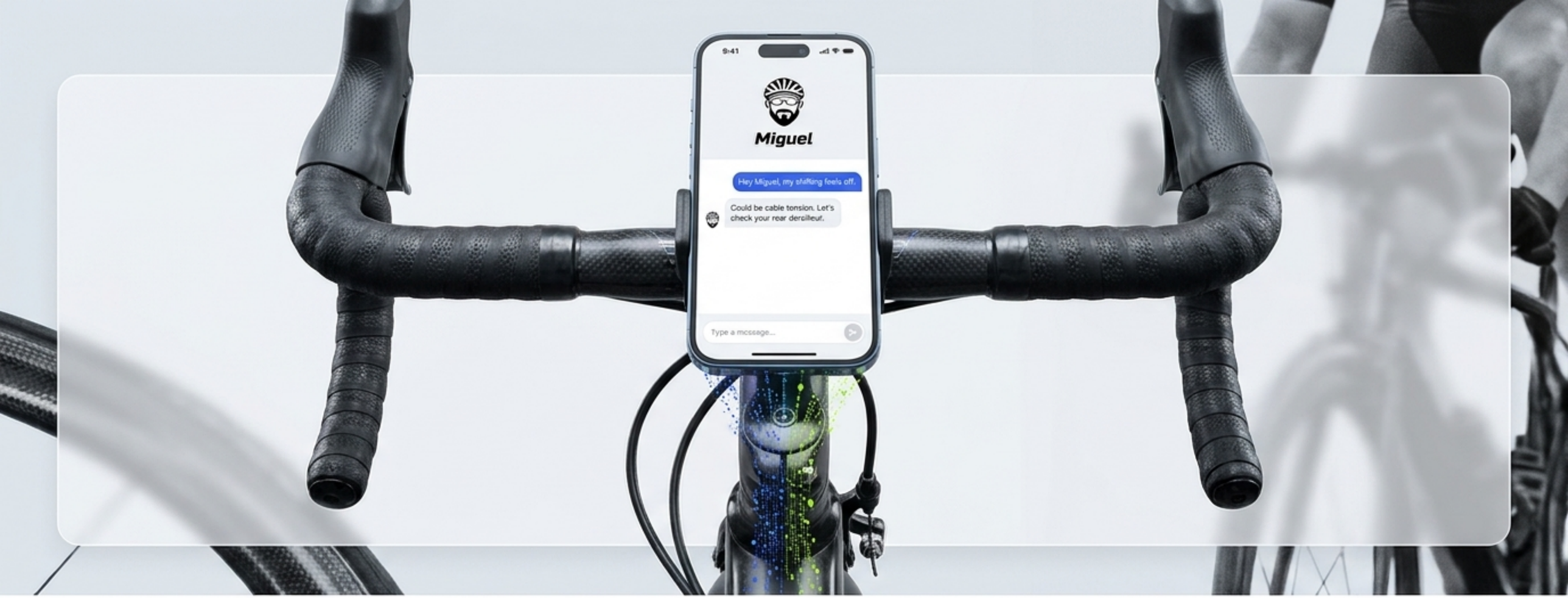
Strategic Insight: 45%





The modern cyclist has the telemetry data, but lacks a centralized, context-aware intelligence to make sense of it all.

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The Solution: Meet Miguel

The first multi-agent conversational AI built **exclusively** for the **cycling world**. Miguel acts as your personal cycling mechanic, race analyst, nutrition guide, and training companion—all rolled into one.

Conversational Intelligence

Ask naturally; no need to speak in rigid technical jargon.

Practical Expertise

Actionable guidance for maintaining and repairing your bicycle.

Contextual Assistance

Remembers your bike, your fitness, and your goals.

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The Engine: Multi-Agent Knowledge Architecture



By segmenting the architecture, Miguel ensures hyper-specific accuracy across the sport's deeply specialized domains.

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Scenario 1: The Mechanic



My gears are skipping when I shift into the larger cogs on a steep climb.

Since it's happening under load on the larger cogs, we need to check your B-tension screw and cable stretch.

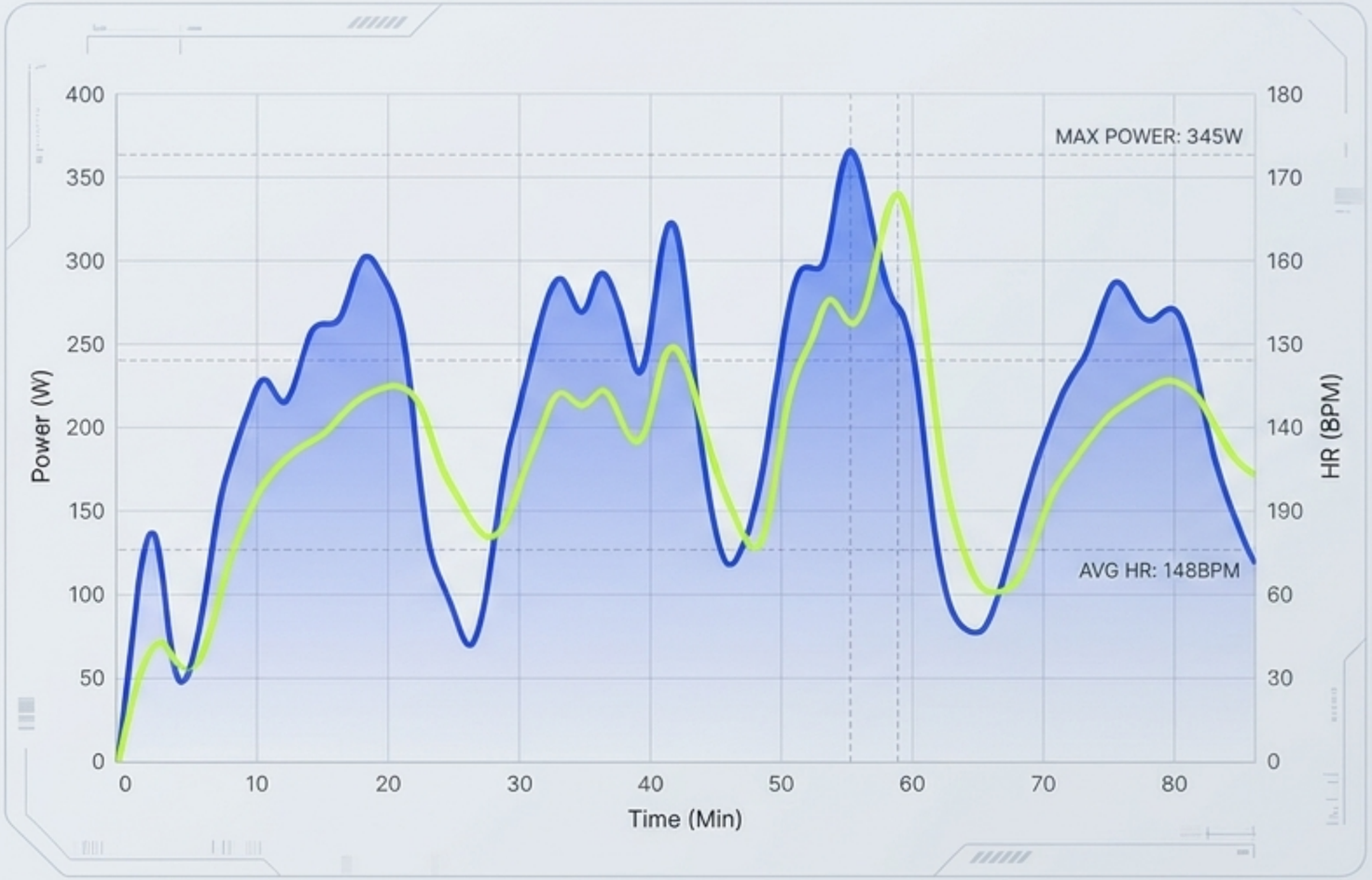
First, shift into your largest chainring and largest cog. Is the upper jockey wheel touching the cassette?

🛠️ [View Shimano B-Tension Guide]

Real-time diagnosing of mechanical problems, minimizing downtime and shop fees.

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Scenario 2: The Physiologist



I have a 100-mile ride tomorrow with 8,000ft of climbing. How should I fuel?

For a century with that much elevation, you'll be burning heavy glycogen. You need 60-90g of carbs per hour. Let's build a hydration and gel strategy based on your Zone 2 Here is your hourly breakdown...



Nutrition Plan

Hourly Target: 1 Bottle + 2 Gels

Translating complex training principles into ride-day execution.

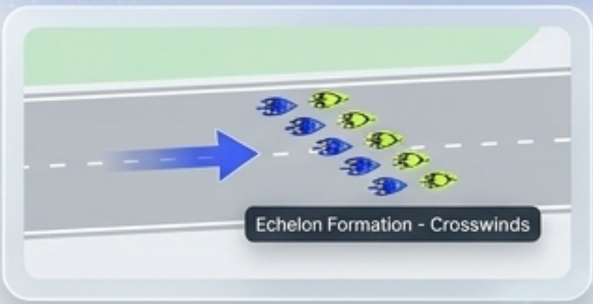
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Scenario 3: The Directeur Sportif



Why did Team Visma ride at the front today if they didn't even want to win the stage?

They were controlling the breakaway's time gap to protect the yellow jersey without "burning matches." By dictating the pace on the front, they also kept their leader safe from crashes in the crosswinds...



Deepening the user's connection to Grand Tours, Classics, and the stories behind the sport.

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More Than a Chatbot: Contextual Intelligence

Contextual Memory Loop

One-month timeline

Interaction 1

User asks about tubeless tires for a 2022 Trek Emonda.

[Miguel logs: Bike = Trek Emonda]

Interaction 2

User asks for a 4-week FTP builder plan.

[Miguel logs: Goal = Power increase]

Today

User asks: “What tire pressure for today’s wet ride?”

High-Energy Cobalt

Based on your Trek Emonda’s 28mm tires and your current FTP training block fatigue, drop it to 65 PSI for better grip.

Miguel builds on previous conversations. You never have to repeat your bike specs, fitness level, or terminology preferences.

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The Knowledge Landscape Matrix

	Miguel	Reddit/Forums	YouTube	Generic AI (ChatGPT)	Local Bike Shop
Instant Accessibility	✓	✗	✗	✓	✗
Cycling-Specific Accuracy	✓	✓	✓	✗	✓
Personalized Context (Memory)	✓	✗	✗	✗	✓
Zero Cost / Friction	✓	✓	✓	✓	✗

Miguel is the only solution that combines the immediate availability of generic AI with the specific, contextual accuracy of a professional mechanic.

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Strategic Expansion Roadmap

Waypoint 1:
Foundation (Current Status)

- Miguel is currently learning!
- Ingesting 8 core knowledge domains.
- Open-Source release under MIT License.
- Community beta and developer contributions.

Waypoint 2:
Integration (Upcoming)

- Mobile app launch (iOS/Android).
- Strava / Garmin API integrations to read live telemetry.
- Advanced personalized training plans.

Waypoint 3:
Ecosystem (Future)

- E-commerce integration (automated replacement part suggestions based on mileage).
- Premium personalized coaching tiers.

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The Holistic Cycling Ecosystem



Cycling is not just an activity; it is a lifestyle that demands expertise across machinery, biology, and culture. Miguel is the only platform designed to bridge the garage, the road, and the television screen.

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Every ride has a story. Every question deserves an answer.

Ride more. Know more. Ride better.

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License: Built for the love of cycling. Open for everyone. (MIT License)

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